CDS540 Group Project

Timeline

• Form your group and report the group members to the TA. (Oct 9)

• Select a Computer Vision project topic and report it to the TA. (Oct 16)

• Submit your group presentation slides and report to Moodle. (Nov 20)

• Deliver your group presentation according to the announced schedule. (Nov 20 & 27)

• Complete and submit your peer evaluation form privately. (Nov 20 & Nov 27)

Candidate Topics

Your project should be related to **Computer Vision**. The following topics are examples; other Computer Vision topics are also welcome.

1. Image processing and enhancement

Image denoising, deblurring, contrast enhancement, super-resolution, or image restoration.

2. Image classification and recognition

Object or scene classification, visual recognition, or visual anomaly detection.

3. Object detection and image segmentation

Object detection, semantic or instance segmentation, text detection, or medical image segmentation.

4. Video understanding and analysis

Object tracking, action recognition, motion estimation, or video event detection.

5. 3D computer vision

Depth estimation, stereo vision, 3D reconstruction, or point cloud processing.

6. Generative models for visual content

Image or video generation, image editing, image-to-image translation, or style transfer.

CDS540 小组项目

时间安排

- 组建小组，并向助教报告小组成员名单。（10月9日）
- 选定计算机视觉项目主题，并向助教报告。（10月16日）
- 将小组汇报幻灯片和报告提交至 Moodle。（11月20日）
- 按公布的日程进行小组汇报。（11月20日和27日）
- 独立、私下完成并提交组员互评表。（11月20日和27日）

候选主题

项目应与计算机视觉相关。以下为示例，也欢迎其他计算机视觉主题。

1. 图像处理与增强

图像去噪、去模糊、对比度增强、超分辨率或图像复原。

2. 图像分类与识别

物体或场景分类、视觉识别或视觉异常检测。

3. 目标检测与图像分割

目标检测、语义分割或实例分割、文本检测或医学图像分割。

4. 视频理解与分析

目标跟踪、动作识别、运动估计或视频事件检测。

5. 三维计算机视觉

深度估计、立体视觉、三维重建或点云处理。

6. 视觉内容生成模型

图像或视频生成、图像编辑、图像到图像转换或风格迁移。

Notes for Selecting Your Topic

Personal Interests

Choose a visual task that interests your team. For example, a photography-related project could enhance image quality, while a sports-related project could track players or recognize actions in videos.

Course Content and Learning Goals

Apply concepts covered in CDS540, such as image processing, convolutional neural networks, vision transformers, detection, segmentation, or visual generation. Topics beyond the course content are welcome as long as they are related to Computer Vision.

Practicality and Feasibility

Consider the available data, computing resources, time, and your team's experience. You may use existing libraries and pre-trained models. Choose a manageable task and clearly explain your implementation, experiments, and findings.

Relevance and Impact

Consider a practical visual problem, such as detecting product defects, analyzing traffic, or improving low-quality images. Define a clear objective and evaluate whether your approach achieves it.

Team Collaboration

Choose a topic that reflects the interests and skills of all group members. Divide the work fairly so that each member makes a meaningful contribution.

Notes for the Group Project Report

Upload the report to Moodle as a **.pdf** file. Use **Times New Roman, 12-point font, 1.25 line spacing, and 0.5-line spacing after each paragraph**. The main text must not exceed **8 pages** of A4 paper in portrait orientation.

Describe the problem, relevant background, method, data, experiments, results, and limitations. Acknowledge any external code, datasets, pre-trained models, and other sources used.

Notes for Group Presentation Slides

Upload the slides to Moodle as a **.ppt, .pptx, or .pdf** file. You must use the slides downloaded from Moodle when presenting.

Notes for the Group Presentation

Each group has **20 minutes** in total: **17 minutes for the presentation and 3 minutes for Q&A**. The presentation order will be random and announced on Moodle before the presentations begin.
Please note: If you exceed the time limit, your presentation will be interrupted.

Notes for Peer Evaluation

Individual scores will be based on group performance and peer evaluation. Each student must privately submit an evaluation of the contributions and performance of the other members of their group.

选题注意事项

个人兴趣

选择团队感兴趣的视觉任务。例如，摄影相关项目可提升图像质量；体育相关项目可跟踪运动员或识别视频中的动作。

课程内容与学习目标

应用 CDS540 涵盖的概念，例如图像处理、卷积神经网络、视觉 Transformer、检测、分割或视觉生成。课程以外的主题也可以选择，只要与计算机视觉相关。

实用性与可行性

考虑可用数据、计算资源、时间和团队经验。可以使用现有程序库及预训练模型。选择工作量可控的任务，并清楚说明实现方法、实验和研究发现。

相关性与影响

考虑实际视觉问题，例如检测产品缺陷、分析交通状况或改善低质量图像。明确目标，并评估所采用的方法是否达成目标。

团队协作

选择能体现全体成员兴趣与技能的主题。公平分工，确保每位成员都作出有意义的贡献。

小组项目报告要求

将报告以 PDF 格式上传至 Moodle。使用 Times New Roman 字体、12 磅字号、1.25 倍行距，段后间距为 0.5 行。正文不超过 8 页纵向 A4 纸。介绍问题、相关背景、方法、数据、实验、结果和局限性。注明所使用的外部代码、数据集、预训练模型及其他资料来源。

小组汇报幻灯片要求

将幻灯片以 .ppt、.pptx 或 .pdf 格式上传至 Moodle。汇报时必须使用从 Moodle 下载的幻灯片。

小组汇报要求

每组共 20 分钟：17 分钟汇报，3 分钟问答。汇报顺序随机决定，并将在汇报开始前通过 Moodle 公布。注意：超时将被打断。

组员互评要求

个人成绩根据小组表现和组员互评确定。每位学生必须私下提交对本组其他成员贡献和表现的评价。